Exercise list R2c — Financial planning and forecasting

Answer key (full derivations in `lista-r2c-financial-planning-forecasting-solutions.pdf`)

1. COGS by percent of sales (Berk 18.1)

$c = 72{,}000/100{,}000 = 0.72$; COGS $= 0.72 \times 110{,}000 = \$79{,}200$.

2. Reading EFN (Higgins 3.1, 3.4)

3.1 Negative EFN = projected cash surplus above the minimum; set the plug to zero and add it to cash (matches p. 405). **3.4** (a) True: $\partial EFN/\partial AR = 1$ since $EFN = TA - (L + E)$. (b) False: same assumptions give the same need. (c) False: an annual forecast is a year-end snapshot; seasonal needs within the year can still arise.

3. Corporação Felipe (Ross 4.1, 4.2)

4.1 Sales 26,450, costs 19,205, NI 7,245; assets 18,170 = debt 5,980 + equity 12,190. Plug = dividends $= 7{,}245 - (12{,}190 - 10{,}600) = \$5{,}655$. **4.2** Retained 3,622.5; equity 14,222.5; debt 5,200; NAF $= 18{,}170 - 19{,}422.5 = -\$1{,}252.5$ (surplus).

4. Growth rates (Berk 18.16)

(a) ROA$\times b = 50/400 = 12.5\%$. (b) ROE$\times b = 50/250 = 20\%$. (c) $20\% \times 0.6 = 12\%$.

5. Universal Products (Higgins 3.5)

(a) $NFA = 2{,}000 + 500 - 260 = 2{,}240$. (b) $E = 2{,}950 + 0.5(156) = 3{,}028$. (c) $EFN = (2{,}750 + 2{,}240) - (1{,}155 + 500 + 3{,}028) = 4{,}990 - 4{,}683 = \$307$ thousand. Matches p. 405.

6. Net new financing (Berk 18.2, 18.3)

18.2 $E_1 = 300{,}000 + 45{,}000 = 345{,}000$, $L_1 = 130{,}000$; NNF $= 500{,}000 - 475{,}000 = \$25{,}000$. **18.3** Debt + equity fund 470,000 at $D/E = 1/3$: $E = 352{,}500$, $D = 117{,}500$; issue \$7,500 equity and \$17,500 debt.

7. João Herdeiro (Ross 4.9–4.11)

4.9 Sales 45,600, costs 22,080, EBT 23,520, tax 7,996.8, NI 15,523.2; dividends 6,240; addition to RE \$9,283.20. **4.10** Cash 8.03%, AR 18.16%, inventory 20.00%, CA 46.18%, fixed 90.79%, TA 136.97%, payables 3.42%; loans, long-term debt, equity: n/a. **4.11** NI 14,876.4, addition to RE 8,896.4; TA $= 59{,}857.5$; L&E $= 8{,}295 + 25{,}000 + 27{,}846.4 = 61{,}141.4$; NAF $= -\$1{,}283.90$.

8. Global Corp. (Berk 18.8–18.11), $ millions

18.8 (a) 189.108; (b) 1.296; (c) NI $= (11.232 - 7.7)(1 - 0.7/2.7) = 2.616$; (d) 25.056; (e) 19.98; (f) 16.524; (g) 122.148; (h) 37.476. **18.9** $E = 22.2 + 0.5(2.616) = 23.508$. **18.10** NNF $= 183.708 - 174.184 = 9.524$. **18.11** $(1 - p)2.616 \geq 1.832 \Rightarrow p \leq 29.98\%$: payout falls from 50% to about 30%.

9. KMS vs. SGR; new debt; IZAX (Berk 18.17–18.19)

18.17 Faster: sales growth 18.0, 16.8, 16.0, 15.3, 14.8% vs. SGR 3.8, 8.2, 8.3, 8.4, 8.5% (2011–2015); book D/E rises from 6% to 32%. **18.18** SGR $= 0.12\times0.75 = 9\%$; new debt $= 0.09\times400{,}000 =$ \$36,000. **18.19** $g_{int} = 0.05 \times 0.75 = 3.75\%$; $D/E = 100{,}000/315{,}000 = 0.3175$ (from 0.3333).

10. Under Armour (Higgins 4.8)

(a) $g^* = PRA\hat{T}$: 14.96, 17.41, 20.06, 20.16, 19.95%. (b) Yes: actual growth (18.1–38.4%) above g^* every year. (c) Higher margin and, mainly, higher leverage; no dividends. (d) $P = 8\%$: 22.8%; $\hat{T} = 2.4$: 25.2%; both: 28.8% ($> 27.1\%$ actual).

11. Turismo da Onça (Ross 4.25–4.27)

4.25 NI 136,760, addition to RE 95,732; TA 679,080, L&E 675,232; NAF = \$3,848. **4.26** Capacity sales $929{,}000/0.8 = 1{,}161{,}250 > 1{,}114{,}800$: fixed assets stay 413,000; NAF $= 596{,}480 - 675{,}232 =$ $-$\$78,752. **4.27** $D/E = 243{,}000/322{,}900 = 0.7526$; debt allowed 315,044; NAF $= 679{,}080 - 418{,}632 -$ $315{,}044 \approx -$\$54,596.

12. R&E Supplies (Higgins 3.9, 3.8)

3.8 `D19 = C19 + C19*D4`; `D37 = C37 + D28`. **3.9** Solve the circular interest: $X = K/(1 - 0.375 \times 0.1)$. (a) $X = 2{,}410.54/0.9625 \approx$ \$2,504 thousand ($\approx$ \$986 thousand above 2022's 1,518). (b) COGS 84%: \$2,243 thousand (-10.4%). (c) Recession: \$2,959 thousand (+18.1%; book +18.2%), with the formula's negative dividend; $\approx$ \$3,006 thousand if dividends are floored at zero. Matches p. 406.

13. Deriving the formulas (Ross 4.31–4.33)

4.31 NAF $= Ag - ML(V)b(1+g) = -ML(V)b + (A - ML(V)b)g$. **4.32** NAF $= 0 \Rightarrow g = ML(V)b/(A - ML(V)b) = \text{ROA}\,b/(1 - \text{ROA}\,b)$; with debt rising at D/P per retained dollar, $A \to P$ gives $\text{ROE}\,b/(1 - \text{ROE}\,b)$. **4.33** With $P_e = P_b + b\,NI$: $\text{ROE}_e b/(1 - \text{ROE}_e b) = b\,NI/(P_e - b\,NI) = b\,NI/P_b$; same for ROA when assets grow only by retained earnings.

14. NAF and growth (Ross 4.28, 4.29)

4.28 NAF$(g) = 412{,}815(1+g) - 491{,}530$: $-16{,}793$ (15%), 3,848 (20%), 24,489 (25%); zero at $g = 19.07\%$ vs. formula 16.16%, because payables grow with sales (18.78%) and interest is fixed. **4.29** NAF$(g) = 416{,}783.86(1+g) - 554{,}736.22$: $-54{,}596$ (20%), $-12{,}917$ (30%), 7,922 (35%); zero at 33.10% vs. formula 32.24%, because interest is fixed (margin rises with sales).

15. KMS at 0.25% share growth (Berk 18.12–18.15, 18.20–18.22), \$000

18.12 Production 1,000; 1,076.25; 1,157.63; 1,244.42; 1,337.05; 1,435.84: expansion needed in 2012. **18.13** Interest 306, 306, 1,666, 1,666, 1,666; tax shields 107.1, 107.1, 583.1 ($\times$3). **18.14** NI 8,038.3; 7,911.2; 8,233.7; 9,557.0; 11,007.4. **18.15** NWC = 39% of sales; increases 2,907.1; 3,118.9; 3,398.8; 3,704.7; 4,040.8. **18.20** With capex 5,000 (2011), 25,000 (2012), 8,000 after: FCF 5,773.1; $-12{,}610.8$; 5,376.9; 6,448.2; 7,610.6. **18.21** $CV = 8.5 \times 26{,}161.5 = 222{,}373$. **18.22** $V = 7{,}995.6 + 222{,}373/1.1^5 + 1{,}540.7 = 7{,}995.6 + 138{,}076.2 + 1{,}540.7 \approx$ \$147.6 million (vs. \$201.0 million in the chapter's 1% scenario).